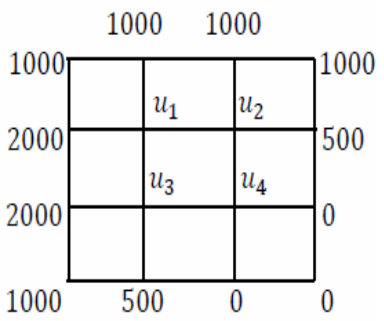
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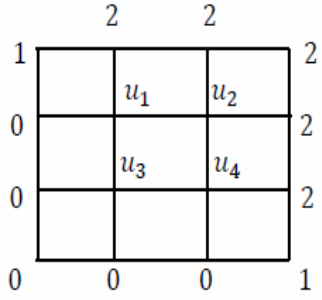
Numerical methods for PDE

Laplace equation

1. Determine the value of $u(x,y)$ satisfying the Laplace equation $\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}=0$ at the pivotal points of a square region with boundary values as shown in the figure. Carry out 3 iterations.



2. Solve $u_{xx}+u_{yy}=0$ in the following square region with the boundary conditions indicated as shown in the figure. Perform 3 iterations.



3. Solve $u_{xx}+u_{yy}=0$ in $0\le x\le 3, 0\le y\le 3$ given that $u(x,0)=0, u(0,y)=0, u(x,3)=x^2$ and $u(3,y)=3y$. Take $h=k=1$ and employ Gauss Seidel method to solve this problem and carry out three iterations by considering the value up to 4 decimals.

4. Solve $u_{xx}+u_{yy}=0$ in $0\le x\le 1, 0\le y\le 1$ given that $u(x,0)=0, u(0,y)=0, u(x,1)=6x$ and $u(1,y)=6y$. Take $h=k=\frac{1}{3}$ and employ Gauss Seidel method to solve this problem. Carry out three iterations by considering the value up to 4 decimals.

Heat Equation

1. Solve the heat equation $\frac{\partial u}{\partial t}=4\frac{\partial^2 u}{\partial x^2}$ subject to the conditions $u(0,t)=0=u(8,t)$ and $u(x,0)=4x-\frac{x^2}{2}$ at the points $x=i: i=0,1,2,\ldots.8$ and $t=\frac{1}{8}j: j=0,1,2,\ldots.5$.

2. Solve the heat equation $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$ subject to the conditions $u(x,0) = \sin \pi x\,, 0 \le x \le 1$ $u(0,t) = u(1,t) = 0$ using Schmidt method. Carry out computations for two levels, taking $h = \frac{1}{3}$ and $k = \frac{1}{36}$.

3. Solve the heat equation $\frac{\partial u}{\partial t} = 2\frac{\partial^2 u}{\partial x^2}$ subject to the conditions $u(0,t) = 0 = u(4,t)$ and $(x,0) = x(4-x)$. Employ Bendre-Schmidt method with $h = 1$ to find the solution of the equation for $0 \le t \le 1$.

4. Solve the heat equation $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$ subject to the conditions $u(0,t) = 0 = u(5,t)$ and $(x,0) = x^2(25 - x^2)$ by taking $h = 1$ for $0 \le t \le 2$ using Bendre-Schmidt method.

Wave equation

1. By using explicit 3 level formula, solve the wave equation $u_{tt} = 4u_{xx}$ subject to the boundary conditions $u(0,t) = u(4,t) = 0, t \ge 0$ and the initial conditions $u(x,0) = x(4-x), \frac{\partial u}{\partial t}(x,0) = 0, 0 \le x \le 4$. Take $h = 1$ and $k = \frac{1}{2}$ an compute $u(x,t)$ for $0 \le t \le 2$.

2. Solve $\frac{\partial^2 u}{\partial t^2} = \frac{\partial^2 u}{\partial x^2}$ with the boundary conditions $u(0,t) = u(1,t) = 0, t \ge 0$ and the initial conditions $u(x,0) = \sin(\pi x)\,, \frac{\partial u}{\partial t}(x,0) = 0, 0 \le x \le 1$ taking $h = k = 0.2$ at $t = 1$.

3. Evaluate the pivotal values of the equation $u_{tt} = 16u_{xx}$ taking $h = 1, k = \frac{1}{4}$ up to $t = 1$. The boundary conditions are $u(0,t) = u(5,t) = 0$ and the initial conditions $u(x,0) = x^2(5-x)$ and $\frac{\partial u}{\partial t}(x,0) = 0$.

4. By using explicit three-level formula, solve the wave equation $4\frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 u}{\partial t^2}$ subject to the conditions $u(0,t) = u(1,t) = 0, t \ge 0, u(x,0) = \begin{cases} \frac{5x}{3}, & 0 < x \le \frac{3}{5} \\ \frac{5(1-x)}{2}, & \frac{3}{5} < x < 1 \end{cases}$ and $\frac{\partial u}{\partial t}(x,0) = 0$. Taking $h = \frac{1}{5}\,, k = \frac{1}{10}$ carry out computations up to $t = \frac{2}{5}$.

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COMPLEX ANALYSIS

1. Show that the following functions are analytic
   (a) F(z) = x + sinx coshy + i(y + cosx sinhy)
   (b) F(z) = $\frac{x-1}{(x-1)^2+y^2}$ -i $\frac{y}{(x-1)^2+y^2}$
2. Prove that analytic function with constant modulus is constant.
3. If $f(z)$ is analytic function , show that
$$\left[\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}\right]|f(z)|^2 = 4|f'(z)|^2$$
4. An electrostatic field in the xy – plane is given by the potential function $\varphi = x^2 - y^2$, find stream function.
5. Prove that $u = x^2 - y^2 - 2xy - 2x + 3y$ is harmonic. Find a function v such that f(z) = u + iv is analytic.
6. Find the orthogonal trajectories of the family of curves $r^2 \cos 2\theta - r\cos\theta + 2 = c$
7. In a two dimensional fluid flow, the complex potential is w(z) = u + iv show that velocity potential $u = e^x(x\cos y - y\sin y)$ is harmonic and find its harmonic conjugate.
8. Find the analytic function $f(z)$ whose imaginary part is $e^x(xsin\, y + y\cos y)$
9. Obtain the power series expansion of the function $f(z) = \frac{z^2-1}{z^2+5z+6}$ in the following regions.
   (i) $|z| < 2$ **(ii)** $2 < |z| < 3$ **(iii)** $|z| > 3$
10. Expand $f(z) = \frac{1}{(z+1)(z+3)}$ in Laurent's series valid for
    (i) $0 < |z+1| < 2$, **(ii)** $|z+1| > 2$
11. Evaluate the following integrals using Cauchy's residue theorem.
    (i) $\int_C \frac{1-2z}{z(z-1)(z-2)}dz$, C: $|z| = 1.5$ (ii) $\int_C \frac{z+4}{z^2+2z+5}dz$, C: $|z+1-i| = 2$
12. Evaluate the integral $\int_C \frac{1}{z^2(z^2+4)}dz$ where C is the circle $|z-2i| = 3$ using Cauchy's residue theorem.